

Procedure Plan — Total Hip Arthroplasty (Posterior Approach)

Surgeon: Dr. M. Chen, MD (synthetic)

Date: 2026-04-30

CPT 27130

Expected duration: 90 minutes

SYNTHETIC PHANTOM — NOT A REAL PATIENT

This document is part of the PreOpReel synthetic-phantom demo case (promo code BUTTERBASE0502 / submission butterbase0502). No real patient data is used. Section pointers (§N.M) map 1:1 to the citations in expected.shotlist.json.

Patient Summary

§2.1 Demographics

65 y/o male, BMI 28.4, height 178 cm, weight 90 kg.

§2.2 Comorbidities

Controlled hypertension; mild osteoarthritis of left knee.

§2.3 Labs & Allergies

CBC, CMP, INR within reference range. NKDA. ASA II.

Anatomical Reference (Posterior Right Hip)

Text-only landmark list. Diagrams omitted from the synthetic phantom plan; Gem (Gemini 1.5 Flash) extracts the AnatomyGraph from this list directly.

§3.1 Acetabulum (right)

Bony socket of the pelvis articulating with the femoral head.

§3.2 Femoral head

Spherical proximal end of the femur seated in the acetabulum.

§3.3 Greater trochanter

Lateral bony prominence — primary reference for osteotomy angle.

§3.4 Lesser trochanter

Medial reference for stem position and leg-length restoration.

§3.5 Sciatic nerve

Posterior to the joint capsule; protected during posterior approach.

§3.6 Posterior capsule

Posterior joint capsule, repaired at closure to reduce dislocation risk.

§3.7 Short external rotators

Piriformis, gemelli, obturator internus — released and reattached.

Surgical Approach — Posterior

§4.1 Position

Lateral decubitus, operative side up, secured with hip positioner.

§4.2 Incision

Curved incision centered over the greater trochanter, extending posteriorly.

§4.3 Layered exposure

Through skin, subcutaneous tissue, fascia lata, and gluteus maximus split.

§5.1 — Skin incision and exposure

Curved skin incision; layered dissection to expose the posterior hip capsule.

Patient-facing narrator line should cite this section (§5.1); Mara blocks any line that drifts into recommending.

§5.2 — Capsulotomy and dislocation

T-shaped capsulotomy; controlled posterior dislocation of the femoral head.

Patient-facing narrator line should cite this section (§5.2); Mara blocks any line that drifts into recommending.

§5.3 — Femoral neck osteotomy

Oscillating-saw osteotomy at planned angle relative to the greater trochanter.

Patient-facing narrator line should cite this section (§5.3); Mara blocks any line that drifts into recommending.

§5.4 — Acetabular reaming and cup placement

Sequential reaming; press-fit acetabular component placed at planned inclination/anteversion.

Patient-facing narrator line should cite this section (§5.4); Mara blocks any line that drifts into recommending.

§5.5 — Femoral canal preparation

Sequential broaching of the femoral canal to size.

Patient-facing narrator line should cite this section (§5.5); Mara blocks any line that drifts into recommending.

§5.6 — Femoral component and trial reduction

Stem inserted; trial head used to confirm stability, length, and offset.

Patient-facing narrator line should cite this section (§5.6); Mara blocks any line that drifts into recommending.

§5.7 — Closure with posterior capsular repair

Capsule and short external rotators repaired through bone tunnels; layered closure.

Patient-facing narrator line should cite this section (§5.7); Mara blocks any line that drifts into recommending.

Surgeon Notes & References

§6.1 Posterior precautions

Avoid hip flexion >90°, adduction past midline, and internal rotation for 6 weeks.

§6.2 Weight-bearing

Weight-bearing as tolerated with assistive device; progress per therapy.

§6.3 Anticipated blood loss

Expected blood loss ~250 mL; tranexamic acid per protocol.

§6.4 References

AAOS Clinical Practice Guideline for Total Hip Arthroplasty (PMID:34567890); Posterior capsular repair reduces dislocation risk (PMID:28456712).

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